Python Program 30

Question

30. Program to print numbers from 1 to N using for loop

Code

```
n = int(input("Enter N: "))
for i in range(1, n + 1):
    print(i, end=" ")
print()
```

Output

```
Enter N: 10
1 2 3 4 5 6 7 8 9 10
```

Python Program 31

Question

31. Program to find the sum of digits of a number

Code

```
num = int(input("Enter a number: "))
total = 0
temp = num
while temp > 0:
    digit = temp % 10
    total = total + digit
    temp = temp // 10
print("Sum of digits =", total)
```

Output

```
Enter a number: 4567
Sum of digits = 22
```

Python Program 32

Question

32. Program to reverse a given number

Code

```
num = int(input("Enter a number: "))
rev = 0
temp = num
while temp > 0:
    digit = temp % 10
    rev = rev * 10 + digit
    temp = temp // 10
print("Reversed number =", rev)
```

Output

```
Enter a number: 12345
Reversed number = 54321
```

Python Program 33

Question

33. Program to print the multiplication table of a number

Code

```
num = int(input("Enter a number: "))
for i in range(1, 11):
    print(num, "x", i, "=", num * i)
```

Output

```
Enter a number: 7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
```

Python Program 34

Question

34. Program to find the factorial of a number

Code

```
n = int(input("Enter a number: "))
fact = 1
for i in range(1, n + 1):
    fact = fact * i
print("Factorial of", n, "is", fact)
```

Output

```
Enter a number: 5
Factorial of 5 is 120
```

Python Program 35

Question

35. Program to print all prime numbers between 1 and N

Code

```
n = int(input("Enter N: "))
print("Prime numbers between 1 and", n, "are:")
for num in range(2, n + 1):
    is_prime = True
    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break
    if is_prime:
        print(num, end=" ")
print()
```

Output

```
Enter N: 30
Prime numbers between 1 and 30 are:
2 3 5 7 11 13 17 19 23 29
```

Python Program 36

Question

36. Program to find the largest number in a series of N numbers entered by the user

Code

```
n = int(input("Enter how many numbers: "))
largest = int(input("Enter number 1: "))
for i in range(2, n + 1):
    num = int(input("Enter number " + str(i) + ": "))
    if num > largest:
        largest = num
print("Largest number =", largest)
```

Output

```
Enter how many numbers: 5
Enter number 1: 45
Enter number 2: 12
Enter number 3: 89
Enter number 4: 34
Enter number 5: 67
Largest number = 89
```

Python Program 37

Question

37. Program to find the sum and average of N numbers

Code

```
n = int(input("Enter how many numbers: "))
total = 0
for i in range(1, n + 1):
    num = int(input("Enter number " + str(i) + ": "))
    total = total + num
average = total / n
print("Sum =", total)
print("Average =", average)
```

Output

```
Enter how many numbers: 5
Enter number 1: 10
Enter number 2: 20
Enter number 3: 30
Enter number 4: 40
Enter number 5: 50
Sum = 150
Average = 30.0
```

Python Program 38

Question

38. Program to print Fibonacci series up to N terms

Code

```
n = int(input("Enter number of terms: "))
a = 0
b = 1
print("Fibonacci series:")
for i in range(n):
    print(a, end=" ")
    next_term = a + b
    a = b
    b = next_term
print()
```

Output

```
Enter number of terms: 10
Fibonacci series:
0 1 1 2 3 5 8 13 21 34
```

Python Program 39

Question

39. Program to check whether a number is an Armstrong number

Code

```
num = int(input("Enter a number: "))
temp = num
digits = len(str(num))
total = 0
while temp > 0:
    digit = temp % 10
    total = total + digit ** digits
    temp = temp // 10
if total == num:
    print(num, "is an Armstrong number")
else:
    print(num, "is not an Armstrong number")
```

Output

Run 1:

```
Enter a number: 153
153 is an Armstrong number
```

Run 2:

```
Enter a number: 123
123 is not an Armstrong number
```

Python Program 40

Question

40. Program to check whether a number is a palindrome

Code

```
num = int(input("Enter a number: "))
rev = 0
temp = num
while temp > 0:
    digit = temp % 10
    rev = rev * 10 + digit
    temp = temp // 10
if rev == num:
    print(num, "is a palindrome")
else:
    print(num, "is not a palindrome")
```

Output

Run 1:

```
Enter a number: 121
121 is a palindrome
```

Run 2:

```
Enter a number: 123
123 is not a palindrome
```

Python Program 41

Question

41. Program to display the multiplication table using while loop

Code

```
num = int(input("Enter a number: "))
i = 1
while i <= 10:
    print(num, "x", i, "=", num * i)
    i = i + 1
```

Output

```
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Python Program 42

Question

42. Program to count the number of digits in a number

Code

```
num = int(input("Enter a number: "))
count = 0
temp = num
while temp > 0:
    count = count + 1
    temp = temp // 10
print("Number of digits =", count)
```

Output

```
Enter a number: 98765
Number of digits = 5
```

Python Program 43

Question

43. Program to demonstrate use of break statement in a loop

Code

```
for i in range(1, 11):
    if i == 6:
        print("Break statement executed at i =", i)
        break
    print(i)
print("Loop terminated")
```

Output

```
1
2
3
4
5
Break statement executed at i = 6
Loop terminated
```

Python Program 44

Question

44. Program to demonstrate use of continue statement in a loop

Code

```
for i in range(1, 11):
    if i % 2 == 0:
        continue
    print(i)
print("Loop finished")
```

Output

```
1
3
5
7
9
Loop finished
```

Python Program 45

Question

45. Program to demonstrate use of pass statement in a loop

Code

```
for i in range(1, 6):
    if i == 3:
        pass  # does nothing, only a placeholder
    print("Number:", i)
print("Loop finished")
```

Output

```
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5
Loop finished
```